JOHN KOCHENSPARGER

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SUMMARY

Electrical engineering graduate with a Secret clearance investigation pending since March 2026. Hands-on production experience includes IEC/EN 61000-4-2 ESD compliance testing, lead-free soldering, cable fabrication, and hardware debugging. Two years in R&D at AMETEK Sunpower on cryocooler systems used in defense, medical, and industrial applications. Co-founded Eye Proxy LLC, contributing PCB design, BLE firmware development, and full mechanical enclosure. Served as Avionics Lead for the Ohio University Rocket Club.

SKILLS

Technical: Soldering (iron, hot-air), Cable fabrication, Fiber Splicing
Design: Altium, Kicad, Multisim, AutoCAD, SolidWorks, Autodesk Inventor
Computing/Programming: C, C++, Python, MATLAB, Git, LaTeX
Platforms: Windows, Linux, MacOS

EDUCATION

Ohio University | Athens, OH — Graduation: May 2026
Bachelor of Science in Electrical Engineering (BSEE)

- Ohio University Rocket Club | *Avionics Lead (Fall 25 – Spring 26)* — Aug 2024 – May 2026
- Institute of Electrical and Electronics Engineers (IEEE) | *Member* — Aug 2024 – May 2026
- Campus Crusade for Christ | *Sound Team (Fall 2025)* — Oct 2023 – May 2026

EXPERIENCE

Eye Proxy LLC | Athens, OH — Oct 2025 – Present
Co-Founder – Production Lead

- Designed a covert dashcam disguised as a parking pass.
- Serving as project manager, leading team coordination.
- Fully designed accelerometer circuit and enclosure housing.
- Led all documentation ensuring a professional level of detail and consistency.
- Produced and presented a working prototype at two expos in April 2026.
- Working towards securing a utility patent.

AMETEK Sunpower Inc. | Athens, OH — May 2024 – Apr 2026
Electrical Engineering R&D Intern

- Developed an app to automate controller manufacturing procedures and tests. Conducted hardware and firmware testing on cryocooler controllers.
- Soldered with lead-free solder and fabricated cables as needed to create/debug test setups for compliance testing.
- Monitored component lifecycles, identified replacements for NFND/obsolescent parts.
- Contributed to test reports, user manuals, technical references, manufacturing procedure documents, parts databases, BOMs.

Asphalt Materials Inc. | Marietta, OH — May 2023 – Aug 2023
Plant Operator Intern

- Assisted a controls engineer in implementing a VFD into a PLC to control asphalt pump flow.
- Created 200+ valve alignment sheets for a large majority of the possible material transfers.
- Optimized the mill room layout to be more space efficient and easier to maintain.
- Assisted plant in daily operations: weighing loads, loading asphalt onto trucks, measuring water parameters.

PROJECTS

Cryotel GT Test-Load | AMETEK Sunpower Inc. Dec 2025 – Apr 2026

- Designed a high-power test load replicating the impedance and operating frequency characteristics of the Cryotel-GT cryocooler for controller validation and testing.
- Characterized cryocooler impedance at max power and designed a 600W equivalent load with appropriate thermal management; selected heatsink and fan configuration using steady-state thermal analysis to prevent overheating.

EP1 | Eye Proxy LLC / Senior Capstone Project Oct 2025 – Present

- Provisional U.S. patent filed (2025), working to secure a U.S. utility patent for a novel covert dashcam disguised as a parking pass, different from traditional dashcams by its unique form factor.
- Managed member coordination, tasking, and scheduling while authoring 90%+ of all documentation.
- Designed custom PCB integrating dual-camera modules, solar charging, motion detection, and wireless connectivity within extreme constraints (8mm thickness, −4°F to 140°F); personally contributed accelerometer circuit (ISM330DHCX via I2C), ESP32-C6 BLE streaming firmware, and full enclosure design.

Rocket Club Flight Controller | Ohio University Rocket Club Oct 2025 – Dec 2025

- Led avionics team component selection based on physical constraints and operational requirements; selected SparkFun Things+ NINA-B306 for telemetry data acquisition.
- Developed firmware to capture and log flight data to SD card for five different L1 rocket launches.
- Built Python post-processing pipeline to analyze 200k–300k lines of sensor data per flight.

CryoTel DS30 Startup Testing | AMETEK Sunpower Inc. Jun 2025 – Aug 2025

- Discovered a set of initial conditions that resulted in operation failure with our production DS30.
- Analyzed events using an oscilloscope to determine the source of the failure.

IEC/EN 61000-4-2 Compliance Testing (ESD) | AMETEK Sunpower Inc. Aug 2024 – Jun 2025

- Used an ESD16 16.5kV simulator for controlled electrostatic discharges (ESD) on full cryocooler systems.
- Repaired/modified PCB to prevent failures.
- Iterated tests until the controller passed a set number of consecutive trials.
- Captured and analyzed events using an oscilloscope to trigger on overcurrents during testing.
- Fully tested two different cryocooler controller systems.

Consolidated Controller Setup Menu App | AMETEK Sunpower Inc. Jun 2024 – May 2026

- Developed Python scripts to validate controller configurations, temperature sensor calibration, interpret data, provide feedback, and log results for QA, RMA, and MRB troubleshooting.
- Built controller setup scripts that auto-detect board revision, firmware version, COM port, and baud rate.
- Delivered the foundation for interactive controller setup menu of automated procedures, attempting to eliminate misconfigurations.

Valve Alignment Sheets | Asphalt Materials Inc. May 2023 – Aug 2023

- Made 200+ procedure documents for different material transfers.
- Instructions contained information on what valves needed to be open and closed.
- Adhered to OSHA standards. Procedures were safe and most efficient.

National Infrastructure Technician Training Workshop | Ohio University May 2023

- Toured with a Fiber Optics installation and maintenance company.
- Gained firsthand experience splicing fiber and interpreting schematics.
- Was professionally advised by a retired Sergeant Major.